

CrackID: A Within-Visit Benchmark for Crack Re-Identification in Ticket Management Systems

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Abstract—Building ticket management systems must determine whether a newly reported defect corresponds to an existing ticket or requires a new one. Repeated reports of the same crack from different viewpoints fragment history and duplicate work. We study this association as crack re-identification and introduce CRACKID (140 photographs, 17 walls, 212 identities in the frozen benchmark snapshot: 6 validation walls and 11 test walls with 19 matchable test identities). We evaluate ten zero-shot crop-based matchers from four families under retrieval, thresholded decisions, and open-set identification. Retrieval varies (Rank-1 0.352–0.654), but validation-calibrated pairwise decisions stay at or below the best-constant floor (F_1^{cv} 0.293–0.351 against a 0.305 floor; oracle F_1^* 0.305–0.359). Context and crack-erasure controls implicate wall location over crack appearance. A geometric baseline registering wall texture with cracks masked out and matching projected masks reaches 0.768 detection-and-identification rate at 0.1 false alarms on 48% of pairs, against 0.429 for the best crop method, 0.404 for the best context control, and 0.000 for a coverage-only control. All figures are point estimates on a small wall-clustered sample with single-annotator homography-propagated labels, so the open-set advantage is provenance-conditioned and ranking is short-baseline within-visit correspondence. This establishes a dataset and benchmark for the visual association step, leaving cross-visit performance and ticket reduction for future work.

Index Terms—crack re-identification, dataset, benchmark, building maintenance, structural inspection, homography, geometric alignment, annotation quality

I. INTRODUCTION

Building ticket management systems organise reports of physical defects into records that support inspection, prioritisation, intervention, and follow-up. A ticket should maintain the history of one incident, but the same crack may be photographed repeatedly by different users or inspections, spawning duplicate tickets that fragment history and duplicate assessment work.

Addressing this requires associating each new observation with its existing ticket, or creating a new ticket for unrecorded damage.

Photographic evidence makes this a visual re-identification problem. Two cases matter: (i) unchanged morphology under varying viewpoint, distance, field of view, illumination and partial visibility; and (ii) physically evolved damage (extension, widening, branching) whose identity must persist despite condition change, distinguishing damage change from acquisition change. Either error creates a duplicate or misattributes the observation.

Existing crack methods address detection and segmentation within single images. Datasets such as CrackSeg9k [1] and OmniCrack30k [2] lack cross-observation identities, so detection cannot link an observation to a ticket. Whether person/object re-identification [3], [4] transfers to cracks—where distinct defects look alike and one defect varies widely—is untested.

We introduce CRACKID for deciding whether observations show the same wall crack: 140 photographs of 17 walls in a frozen snapshot (6 validation, 11 test; 19 matchable test identities, $n_Q=280$). Real photos cover acquisition variation within one visit; a synthetic protocol preliminarily probes appearance change and growth. Real cross-visit performance remains future work. All accuracy figures are point estimates on a small wall-clustered sample.

Our benchmark tests ten matchers zero-shot on CRACKID, separating candidate retrieval from thresholded same-crack decisions and open-set identification with rejection: ranking a record highly does not justify attaching a report to it. Context and crack-erased controls separate crack cues from wall location, and a geometric baseline registers walls and compares projected masks.

The study asks RQ1 (association under acquisition change), RQ2 (roles of context and geometric alignment), and RQ3 (behaviour under synthetic appearance/growth change). RQ1–RQ2 use real single-visit photos; RQ3 is exploratory synthetic. Ticket-level duplicate reduction and cross-visit evaluation are left for future work.

Contributions: (i) CRACKID with cross-observation identities and stated annotation provenance (Sec. III-B); (ii) a benchmark separating retrieval, same-defect decisions, and open-set identification (Sec. IV, VI); (iii) context controls and a geometric reference with coverage analysis (Sec. V, VI-C, VII-B); (iv) a preliminary synthetic revisit probe (Sec. VII-D).

II. RELATED WORK

Crack analysis targets detection and segmentation [5]–[7]; its datasets annotate single images and consolidations inherit that scope [1], [2], and multi-view pipelines fuse one inspection into a surface or segmentation [8] without instance correspondence. Structural-health monitoring compares cracks over time with related machinery [9]–[12], but identity is *given by construction* (fixed camera, marked location): they measure a known crack rather than deciding association under an open-set protocol, which is our target.

Object re-identification [4] with open-set identification [13] is our formal setting; keypoint matching [14], [15], learned correspondence (SuperGlue [16], LoFTR [17]), and region embeddings [3], [18]–[21] apply to crack crops, all benchmarked here zero-shot without CRACKID fine-tuning (YOLOv8 row: layer-9 `yolov8n.pt` features [22]; LightGlue [23] and NetVLAD [24] left as future retrieval baselines). Vehicle-damage re-identification across claims [25] mirrors our open-set protocol; the rest of that literature covers detection/severity rather than association [26]–[28]. Aligning captures then reasoning about presence versus absence is change detection [29], [30] with classical planar registration [31]–[33] and ECC fallback [34]; our point is that alignment and matching targets should be *disjoint*, with the homography driven by the wall because the crack is what is to be identified.

III. DATA ACQUISITION

A. Acquisition and composition

The CRACKID frozen benchmark snapshot holds 140 photographs of 17 exterior walls at one site over four half-days (two consumer handsets; archived at Mendeley Data). Walls span stippled render (textured) and smooth painted plaster (often crack-only in frame). Split: 6 validation walls wall01–wall06 (66 photos, 66 identities) and 11 test walls wall07–wall17 (74 photos, 146 identities); only identities in ≥ 2 photos yield answerable queries (18 val, 19 test; $n_Q=280$ test, 439 val). Two walls added later (wall18–wall19) are excluded from every number in this paper. Each wall is one continuous walk-around (median 3 s between consecutive frames, max span 81 s), so the data measures within-visit viewpoint/scale correspondence, a hypothesized upper bound on multi-visit performance.

Adjacent-frame retrieval. 100% of 280 answerable test queries are answered by the adjacent frame; Sec. VII gap sweeps (280→205→149→105 for gaps 0→3) find flat Rank-1 on a shrinking pool, characterising short-baseline robustness rather than proving irrelevance.

B. Annotation, and why it needs auditing

One click per crack is propagated across consecutive frames via ORB homography plus one-to-one assignment, then fragment merges join components of one physical crack. Merge risk concentrates here: a wrong join turns a hard negative into a free positive for *every* method. We report extent, not correctness: 95/212 identities (45%; 70/146, 48% on test) come from merges, with median bridged gaps of 228 px (test) and 286 px (val). A single annotator validated all clicks with full triage of unclaimed points (107/1 023 naming no crack; full counts in supplement); agreement is unmeasured, so merge risk is bounded by reporting, not removed. Counts refer to the frozen snapshot (Table I); the live `dataset/labels/` has since been re-edited (wall18–wall19 added, changelog rebuilt without gap fields) and no longer reproduces these counts until the benchmark is re-run.

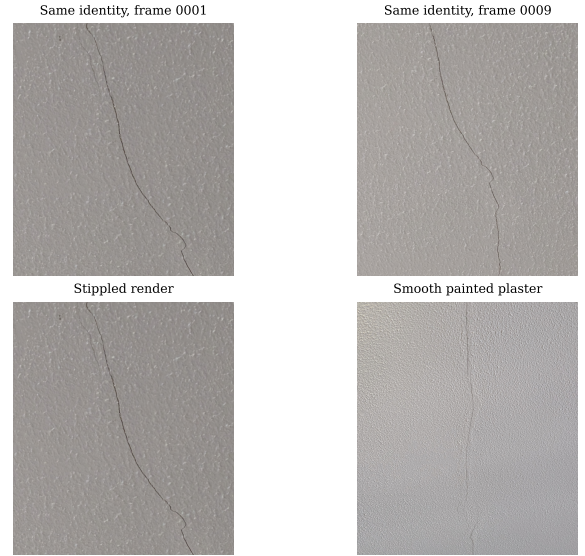


Fig. 1. Real CRACKID captures. Top: one identity in non-adjacent frames. Bottom: stippled render vs smooth plaster. Labels use homography propagation; illustrative, not an independent audit.

IV. BENCHMARK PROTOCOL

The benchmark measures the visual association component: given a new crack observation, decide whether it matches a recorded defect or has no match in the reference set. Annotated identities serve as proxies for persistent defect records; we measure association accuracy, not duplicate-ticket reduction or full workflow performance.

A. Task formulation and candidate records

A *wall* is one physical wall; an *identity* links observations attributed to the same crack (provenance in Sec. III-B). A *query* is one crack instance; its *gallery* holds eligible instances from other photographs of the same wall, assuming the wall is known. Cross-wall search is out of scope. A query is *known* if its gallery contains the same identity, else *unknown*. Each method scores query–gallery pairs for ranking, thresholded decisions, and identification with rejection. A *query–gallery pair* is the pairwise unit; an *image pair* is the registration/assignment unit. The Hungarian one-to-one constraint applies per image pair only; several observations may share one defect record.

Test counts and prevalence are summarised in Table I for the frozen benchmark snapshot (6 validation walls wall01–wall06; 11 test walls wall07–wall17; wall18–wall19 added later are excluded from all numbers). Registration is attempted for all 301 same-wall test image pairs; 255 yield scored pairs with validity-mask cells (see Sec. VII-B).

B. Unchanged damage under different acquisition conditions

The primary evaluation uses real single-visit walk-arounds per wall, where morphology is treated as approximately unchanged and variation comes from viewpoint, distance, scale, illumination, and visibility, with same-wall distractors and

TABLE I
COUNTING UNITS FOR EVERY METRIC (FROZEN SNAPSHOT, $n_Q=280$
TEST).

	Validation	Test
Walls / photographs / identities	6 / 66 / 66	11 / 74 / 146
Identities in ≥ 2 photographs	18	19
Answerable queries	439	280
Positive query-gallery pairs	26 012	5 722
Valid query-gallery pairs	89 118	31 822
Same-wall image pairs	394	301
Scored image pairs (with valid cells)	341	255

unknown queries included. Walls are disjoint across validation/test. All methods are used as-is without CRACKID fine-tuning on shared masks (Sec. V); thresholds are calibrated on validation and applied to test. This establishes within-visit correspondence only, not cross-visit or progression robustness.

C. Controlled appearance change and damage progression

A complementary synthetic revisit isolates photometric change from structural growth: ten source photographs from five walls, one seed, same modest affine warp in both conditions. Condition B corrupts appearance (blur, noise, contrast/brightness) with structure intact; condition A extends one tip of the largest component. Full parameters are in code. This compares OSNet with context against HYBRID-SKEL (geometric alignment plus skeleton fallback/blend, $\tau=8.0$ px), which is distinct from the primary REG+CHAMFER reference. With one seed and five walls, results are preliminary checkpoints, not longitudinal evidence.

D. Decision rules and handling of unavailable scores

Pairwise classification accepts a pair at threshold; query-level identification accepts the top gallery instance only at threshold, else rejects; assignment applies the Hungarian maximisation per image block then rejects below the validation threshold. Unscorable pairs are retained: predicted negative for pairwise/assignment F_1 , rank-last for retrieval. A miss on a known query and a correct rejection on an unknown query are thus distinguished from a supported non-match: registration failure is unresolved and may need review, not proof of a new defect. For assignment $F_1=2TP/(2TP+FP+FN)$, correct accepted assignments are true positives, incorrect accepted ones false positives, unmatched known instances false negatives.

E. Evaluation metrics and reference baselines

Retrieval over known queries uses Rank-1, Rank-5, mAP. Pairwise decisions use precision/recall/ F_1 , distinguishing oracle F_1^* from validation-calibrated F_1^{val} . Assignment uses F_1^a . Identification with rejection uses DIR (known accepted-correct) and FAR (unknown accepted); DIR@FAR= 0.1 is the constrained operating point, distinct from the fixed-threshold result. Coverage is the scored-pair fraction and is reported alongside accuracy. References are random ranking and the

best-constant pairwise F_1 of $2p/(1+p)$ (0.305 at test prevalence 0.180; 0.544 on the 0.48 geometric subset; 0.452 at validation prevalence 0.292); full chance levels in Table II. Resampling is by wall given shared capture conditions; with 11 test walls and 19 matchable identities we report point estimates only, with wall-level intervals and score-matrix release as future work (Sec. VIII).

V. A GEOMETRIC REFERENCE BASELINE

The baseline tests whether crack location on the wall supports association: it aligns wall surfaces to a common frame and compares projected crack masks. It assumes an approximately planar wall and uses no CRACKID training. Shared UNet++ masks (trained on public crack data only; no CRACKID wall in training) with 200 px / 5 px instance extraction keep inputs consistent across methods (full segmenter settings in code). Masks are treated as given; their error is not quantified.

A. Registration using wall texture

For I_a, I_b , a homography H ($x' \sim Hx$) is estimated from surrounding wall texture with detected cracks dilated and excluded, so alignment is driven by surface features independently of the identity under test. This independence is conditional on the masks and stage, not guaranteed per pair. Full detector settings (SIFT with CLAHE contrast enhancement [35]/ratio/robust estimation) are in code.

B. Registration cascade and validity checks

Textured render usually supplies background features; smooth plaster may not. A three-stage cascade is used: masked SIFT, unmasked-SIFT fallback if too few keypoints remain (flagged per pair since the crack then aids alignment), and ECC photometric fallback. Candidates pass plausibility checks (no mirrored/folded quads, scale within $[1/16, 16]$); the surviving stage is recorded per pair and its inlier RMSE sets the comparison tolerance. Unscored pairs are unresolved lack of evidence, not proof of different defects.

C. Crack projection and geometric comparison

Accepted registrations project instances from I_a into I_b for comparison by symmetric mean Chamfer distance, which tolerates small misalignments that would collapse IoU on narrow cracks. Let $d(p, b) = \min_{q \in b} \|p - q\|_2$:

$$D(a, b) = \frac{1}{2} \left[\frac{1}{|a|} \sum_{p \in a} d(p, b) + \frac{1}{|b|} \sum_{q \in b} d(q, a) \right], \quad (1)$$

with early rejection of distant candidates. Chamfer coverage (fraction within tolerance τ from per-pair RMSE, clamped $[2, 16]$ px, default 6 px; HYBRID-SKEL fixes 8.0 px) is evaluated separately as the scale-free counterpart (Sec. VII-C).

D. Correspondence decisions and limits of spatial evidence

Hungarian one-to-one assignment [36] per image pair plus validation threshold (evaluation, identical for all methods); unscored pairs receive $-\infty$ (predicted negative / rank-last per Sec. IV). An aligned projection into visibly empty wall is a **licensed negative**—positive evidence of absence, stronger than low embedding similarity—while missed detections, out-of-frame regions, and registration failures stay unresolved. REG+CHAMFER targets unchanged damage; progression uses HYBRID-SKEL. Performance is conditional on surface-dependent failures and homography-based annotation provenance.

VI. THE APPEARANCE CEILING

A. Setup

Ten methods operate on crack-centred crops under a shared validity mask: SIFT/ORB with RANSAC inliers, SuperGlue/LoFTR with pretrained indoor configs, five embeddings plus re-ID OSNet (cosine similarity, all zero-shot on CRACKID; YOLOv8 is layer-9 `yolo8n.pt` features). Full input sizes and checkpoints are in code. Context controls use OSNet and CLIP at fixed backbone: `@ctx1` expands the instance box by $1\times$ per side, `@ctx1@nocrack` inpaints the crack in the same view, isolating wall context. REG+CHAMFER registers walls and compares projected masks (Sec. V); a COVERAGE-ONLY control scores registered pairs constant without inspecting cracks. The synthetic protocol compares OSNet with context against the distinct HYBRID-SKEL configuration. LightGlue and NetVLAD are left as future retrieval baselines.

B. Result: appearance buys almost nothing on the decision

Table II summarises the primary benchmark ($n_Q=280$ answerable test queries over 11 walls; frozen snapshot). Rank-1 spans 0.352–0.654 (ViT-B/16 and OSNet tied best; OSNet best mAP at 0.520; SuperGlue best crop DIR@FAR= 0.1 at 0.429), yet oracle F_1^* spans only 0.305–0.359 (+0.000 to +0.054 over the 0.305 floor), and validation-calibrated $F_1^{@v}$ spans 0.293–0.351 with two rows below the floor (DeiT, DINOv2); ranking does not transfer to deciding. The post-calibration collapse is consistent with threshold instability on a small wall-clustered sample compounded by a validation-to-test prevalence shift (0.292→0.180; floors 0.452→0.305); the threshold-free check (pairwise AP/AUROC) awaits the score-matrix release (Sec. VIII). With 19 matchable test identities, no fine ranking among crop rows is claimed. All figures are point estimates.

C. Is it the crack, or the wall behind it?

Context helps ranking and open-set but erasure keeps most of it: OSNet Rank-1 0.654→0.775 (0.711 erased), DIR@FAR 0.168→0.404 (0.393 erased); CLIP open-set 0.143→0.311 (0.286 erased). Both erased views still beat every tight crop: the cue is wall location. Pairwise F_1^* stays in the narrow band 0.307–0.361 across base, context, and erased views while

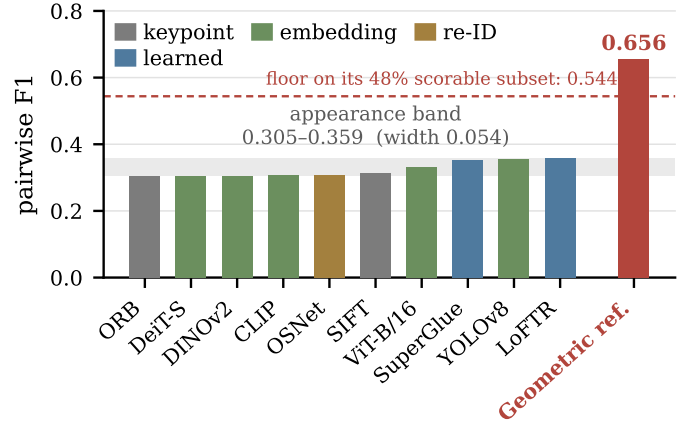


Fig. 2. The appearance ceiling: pairwise F_1^* stays at the 0.305 floor while ranking spreads 0.30. Geometry (red) leaves the band but on the 0.48 pool (own floor 0.544). Point estimates on $n_Q=280$; Table II takes precedence.

open-set moves +0.236 for OSNet, so (RQ1/RQ2) methods retrieve within-visit observations but reliable automatic identity decisions remain limited; a ranked list may assist an operator while acceptance needs further evidence.

VII. RESULTS AND DISCUSSION

A. Main comparison and the open-set gap

Geometric Rank-1 reaches 0.952 vs crop best 0.654 (ViT-B/16 and OSNet tied) and mAP 0.886 vs 0.520 (OSNet) (*scored* 0.48 vs 1.00). Conditioning on coverage as excess over COVERAGE-ONLY, ranking gains stay large, but the open-set number is what coverage cannot explain: DIR@FAR=0.1 of 0.768 vs 0.429 (SuperGlue), 0.404 (best context OSNet+ctx1), 0.038 chance and 0.000 for COVERAGE-ONLY. Crop similarities on self-similar hairlines rarely support rejection, while an aligned empty projection gives a licensed negative. Compute does not explain the gap (LoFTR 8.64s vs OSNet 0.05s per query; Reg 27.9s per query, 26.0s per image pair; timings relative only). RQ2 (geometric part): alignment adds identification beyond coverage alone, conditional on success, surface, and provenance. All numbers are point estimates on 19 matchable identities.

B. Partial coverage, gap structure, and failures

Crop methods score 100% of pairs, geometry 48%; unscored pairs are charged negative/rank-last. Censoring penalises ranking but subsidises pairwise F_1 : the scorable subset keeps 100% of positives but only 36.8% of negatives (prevalence 0.180→0.373, floor 0.305→0.544). COVERAGE-ONLY—constant on registered pairs without inspecting cracks—thus scores 0.544 pairwise F_1 , 0.533 mAP and 0.515 Rank-1: much of the ranking/pairwise margin is *which pairs align*. The open-set lead (0.768 vs 0.000) is the exception we lead with. Failures are wall-dependent (two test walls 0%; full per-wall table in supplement), so aggregates are not uniform across surfaces. All 280 answerable queries are answered by the adjacent frame, so headline ranking is short-baseline

TABLE II

MAIN RESULT ON CRACKID TEST ($n_Q=280$ OVER 11 WALLS; FROZEN SNAPSHOT). F_1^* IS ORACLE-THRESHOLDED ON TEST; ΔF_1^* IS EXCESS OVER THE FULL-POOL 0.305 FLOOR; $F_1^{\text{@v}}$ USES THE VALIDATION-CALIBRATED THRESHOLD. BOLD: BEST CROP ROW PER COLUMN. THE GEOMETRIC ROW IS SCORED ON A 0.48 SUBSET WHOSE OWN FLOOR IS 0.544, SO ITS ΔF_1^* IS NOT COMPARABLE TO CROP ROWS; SEE SEC. VII-B. SCORING, CHANCE LEVELS, AND COSTS: SEC. IV–VII-B. $S/\text{QUERY} = \text{TOTAL}_S/280$ ON ONE WORKSTATION (RELATIVE TIMINGS ONLY).

Method	Family	R@1	mAP	DIR@FAR=0.1	F_1^*	ΔF_1^*	$F_1^{\text{@v}}$	scored	s/query
<i>Reference level (same 0.48 pool as geometry)</i>									
COVERAGE-ONLY	geometry	0.515	0.533	0.000	0.544	0.239	0.544	0.48	—
<i>Crop scope</i>									
ORB [15]	keypoint	0.352	0.364	0.018	0.305	0.000	0.305	1.00	0.038
SIFT [14]	keypoint	0.432	0.417	0.036	0.313	0.008	0.305	1.00	0.086
SuperGlue [16]	learned	0.562	0.455	0.429	0.353	0.048	0.305	1.00	5.826
LoFTR [17]	learned	0.549	0.440	0.336	0.359	0.054	0.351	1.00	8.639
YOLOv8 backbone [22]	embedding	0.561	0.453	0.157	0.357	0.052	0.308	1.00	0.217
ViT-B/16 [18]	embedding	0.654	0.511	0.286	0.331	0.026	0.317	1.00	0.193
DeiT-S [19]	embedding	0.621	0.497	0.250	0.305	0.000	0.293	1.00	0.058
CLIP ViT-B/32 [20]	embedding	0.604	0.471	0.143	0.307	0.002	0.306	1.00	0.050
DINOv2 [21]	embedding	0.607	0.484	0.307	0.305	0.000	0.297	1.00	0.011
OSNet [3]	re-ID	0.654	0.520	0.168	0.309	0.004	0.309	1.00	0.053
<i>Full-image scope</i>									
REG+CHAMFER (reference)	geometry	0.952	0.886	0.768	0.656	—	0.582	0.48	27.92

correspondence; gap sweeps (280→105 at gap 3) leave Rank-1 flat on a shrinking pool rather than proving irrelevance. Registration follows image sharpness over frame gap (wall-clustered Wald $z=5.11$, $p=3.3 \times 10^{-7}$ vs -2.78 , $p=5.4 \times 10^{-3}$; details in supplement); one sharpness gate (≥ 10) raises registration to 56.1% on the filtered set at the cost of one wall, suggesting capture-quality guidance, not full-population coverage.

C. Ablations and cost

Coverage (scale-free) costs 0.240 mAP and 0.103 Rank-1 versus Chamfer distance, with DIR nearly unchanged (+0.014): the comparison rule moves ranking, not decisions. Disabling crack exclusion leaves ranking/pairwise unchanged but raises DIR by +0.057; masking is retained for evidence independence, not observed accuracy here. ECC adds $\sim +0.03$ Rank-1 / +0.02 mAP with DIR unchanged. Cost is 26.0 s per image pair (7817.3 s over 301 test pairs, i.e. 27.9 s per query; Chamfer-only thereafter), about $3 \times$ LoFTR per query and more than two orders above the cheapest embedding; absolute timings are workstation-dependent and SuperGlue/LoFTR were timed in a separate run.

D. A synthetic revisit protocol

Ten source photographs from five walls (one seed) are warped identically with (A) single-tip growth or (B) photometric corruption, ranked against same-wall real instances (full recipe in code; warp modest vs real p90 tilt 75°). HYBRID-SKEL (alignment plus skeleton fallback/blend) differs from Table II REG+CHAMFER, so rows test controlled change, not the main configuration.

OSNet+ctx dominates ranking/open-set (0.826/0.671 and 0.739/0.663 vs 0.275/0.274 and 0.343/0.342), while HYBRID-SKEL holds assignment F_1^a (0.646/0.640 vs

Protocol	Method	n_q	R@1	mAP	DIR@FAR _{0.1}	$F_1^{\text{@v}}$	F_1^a
A (growth)	HYBRID-SKEL	69	0.275	0.401	0.343	0.569	0.646
	OSNet+ctx	69	0.826	0.647	0.739	0.601	0.631
B (appear.)	HYBRID-SKEL	82	0.274	0.403	0.342	0.574	0.640
	OSNet+ctx	82	0.671	0.570	0.663	0.594	0.609

TABLE III

HYBRID-SKEL VS OSNET+CTX ON SYNTHETIC REVISIT; F_1 AT VALIDATION THRESHOLD. ONE SEED, FIVE WALLS, NO RESAMPLING: CHECKPOINT, NOT STABILITY CLAIM. HYBRID-SKEL $\neq$ TABLE II REG+CHAMFER.

0.631/0.609) with pairwise $F_1^{\text{@v}}$ close (0.569/0.574 vs 0.601/0.594); no assignment cell came from pure homography alone ($H\%=0.0$). Ordering thus reverses with metric (RQ3): the Table II advantage does not imply multi-visit robustness, and single-seed controlled growth does not establish real progression performance.

VIII. VALIDITY, LIMITATIONS, AND FUTURE WORK

This work establishes a first measurable formulation of within-visit crack re-identification and a geometric reference baseline. The results are point estimates on a small, wall-clustered sample with single-annotator labels and partial registration coverage, so each finding below is paired with the concrete step that would confirm or extend it.

- 1) **Scale to multi-wall, multi-visit data.** Collect more walls, surface families, and annotators to support wall-level confidence intervals and reliable method ranking. Most importantly, add a second capture session at deliberately varied distances and lighting to convert the within-visit measurement into a genuine cross-visit benchmark for ticket management backends.
- 2) **Add provenance-free annotation.** All clicks were human-validated by a single annotator with full triage of

unclaimed points, but agreement is unmeasured and the geometric advantage remains provenance-conditioned. A small subset labeled entirely without homography propagation, plus released per-pair score matrices for sensitivity analysis, wall-clustered intervals, and threshold-free AP/AUROC, would quantify how much of the open-set gap is independent of annotation provenance.

- 3) **Close the registration coverage gap.** The baseline answers 48% of pairs (97/301 image pairs; 15 328/31 822 valid cells) and scores zero on smooth-plaster walls. Surface-adaptive registration — learned local features, direct photometric alignment, or sparse depth from consumer devices — would push coverage toward full-set evaluation and further decouple the geometry pipeline from homography-based provenance.
- 4) **Learn from geometry-derived supervision.** The baseline is a non-learned reference, not a deployment target. Using the registration pipeline as a teacher for pseudo-labels, or fine-tuning a foundation-model backbone on homography-derived correspondence, would retain the location-based advantage on pairs registration cannot answer.
- 5) **Evaluate the ticket lifecycle.** Define task-level metrics — duplicate rate and missed-new-defect rate — and evaluate segmentation, re-identification, and ticket assignment jointly in a closed-loop simulation of real inspection workflows.

Together, these steps turn the present within-visit reference into a deployable multi-visit re-identification component for operational ticket management.

IX. CONCLUSION

We built a dataset that makes crack re-identification testable: CRACKID, 140 photographs over 17 walls with 212 identities in the frozen snapshot (6 validation walls, 11 test walls; wall18–wall19 excluded), its annotation provenance (45% merged; 48% on test) stated rather than assumed, with the triage of unclaimed clicks complete (107/1 023 naming no crack). All accuracy rows are point estimates with no wall-level intervals (19 matchable test identities, $n_Q=280$); we claim the pattern, not the digits. The benchmark’s main finding is negative and structural: on the decision a record requires, ten zero-shot matchers across four families buy between +0.000 and +0.054 over the 0.305 best-constant floor (F_1^* 0.305–0.359), with eight of ten rows at or below the floor at the validation threshold ($F_1^{@v}$ 0.293–0.351; LoFTR alone at +0.046). Ordering a gallery and deciding an identity are not the same capability. The difficulty is not crack appearance alone: a geometric reference baseline that registers on wall texture with cracks masked out and Chamfer-matches projected masks reaches DIR@FAR=0.1 of 0.768 against 0.429 for the best crop method and 0.404 for the best context method while answering 48% of pairs, and a COVERAGE-ONLY control exploiting that alignment without examining a crack scores 0.000 on the open-set metric (0.544 pairwise F_1 on its own 0.544 floor). Operationally, the open-set advantage

comes from an aligned pair supporting a *licensed negative*, positive evidence of absence, where a crop embedding can only report low similarity; whether that advantage is independent of homography-based annotation remains unresolved and must be read as provenance-conditioned. Two bounds remain addressable: a real second capture session turns the benchmark from viewpoint to cross-visit, and a second annotator plus a homography-free audit subset would turn the single-annotator triage into a measured agreement rate. The 48% coverage is also a deployment gap in its own right, not merely a scoring issue: smooth-plaster walls can be categorically unanswerable. Finally, the small single-seed synthetic revisit checkpoint reverses the headline ranking and open-set result in favour of OSNet+context, while the distinct HYBRID-SKEL pipeline retains only a narrow assignment- F_1 advantage; the present paper therefore establishes a promising within-visit geometric reference, not a solved re-identification component for a multi-visit ticket management backend.

ACKNOWLEDGMENT

This work was supported by “R2UTechnologies — modular systems” (C644876810-00000019), by “R2UTechnologies” Consortium, cofinanced by NextGenerationEU, through the Incentive System “Agendas para a Inovação Empresarial” (“Agendas for Business Innovation”), within the Recovery and Resilience Plan (PRR).

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